

```
1 function v = rocket_velocity(t)
2
3     v = zeros(size(t));
4
5
6     idx1 = (t >= 0) & (t < 8);
7     v(idx1) = 10 * t(idx1).^2 - 5;
8
9     idx2 = (t >= 8) & (t < 16);
10    v(idx2) = 624 - 3 * t(idx2);
11
12    idx3 = (t >= 16) & (t < 26);
13    v(idx3) = 36 * t(idx3) + 12 * (t(idx3) - 16).^2;
14
15
16    idx4 = (t > 26);
17    v(idx4) = 2136 * exp(-0.1 * (t(idx4) - 26));
18 end
19
```